Byeonghwa Jeong

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Education

Ph.D. **University College London**, Geography — London, UK, Sept 2019 – June 2023
- Thesis: An Integrated Deep Learning Model using Retail, Residential and Footfall Changes data to Predict Retail Gentrification.
- Supervisors: Prof. Paul Longley and Prof. James Cheshire.
- Developed an Autoencoder-LSTM model to predict gentrified high streets across England using longitudinal residential, retail, and footfall variables.
- Estimated the long-term impact of COVID-19 on retail gentrification using multivariate multiple regression and likelihood ratio tests.

M.Sc. **The University of Edinburgh**, Geographical Information Sciences — Edinburgh, UK, Sept 2017 – Aug 2018
- Dissertation: Anticipation of Commercial Gentrification through Time and Space: Seoul, South Korea.
- Predicted retail gentrification in Seoul using Self-Organising Maps and a Cellular Automata Markov model.

B.A. **Kyungpook National University**, Geography — Daegu, South Korea, Mar 2008 – Aug 2015

Experience

Oak Ridge National Laboratory, R&D Associate — Knoxville, USA, Feb 2025 – present, 1 year 5 months

Research on high-resolution population modeling, GeoAI, disaster impact assessment, socioeconomic mapping, and global geospatial data products.
- Participating in LandScan Global and LandScan Mosaic projects that generate 1 km and 100 m global gridded population data.
- Integrating satellite-based damage assessments with population models to estimate populations affected by disasters and conflicts.
- Developing a global habitable probability mask using AI to improve gridded population data.
- Building AI models to predict building-level wealth indices from satellite images, building features, and building footprint data.
- Applying multiple instance learning to survey-based data for small-area estimates of household wealth distribution.

University of Toronto, Post-Doctoral Research Fellow — Toronto, Canada, July 2023 – Jan 2025, 1 year 7 months

Research on downtown recovery, footfall analytics, urban spatial data, and machine learning-based analysis of post-pandemic urban change.
- Measured the recovery of downtown areas from the COVID-19 pandemic.
- Estimated uncertainty in sample footfall data and imputed missing footfall data using deep learning-based models.
- Analyzed building-level recovery rates and their relationships with building characteristics.
- Developed downtown boundaries using unsupervised image segmentation and open-source spatial big data.
- Explored socioeconomic determinants of downtown recovery through explainable AI.

Daejeon Sejong Research Institute, External Collaborative Researcher — Daejeon, South Korea

Research project on big data-driven strategies for enhancing the Science City Daejeon brand. — Oct 2024 – Dec 2024, 3 months

- Crawled web content about the Korea Science Festival and Daejeon Science Festival between 2019 and 2024.
- Performed text mining using TF-IDF, Latent Dirichlet Allocation, and network analysis.
- Explored changes in public perceptions of science festivals.

Daejeon Sejong Research Institute, External Collaborative Researcher — Daejeon, South Korea

Research project on discovering regional assets using big data analysis and civic collaboration. — Aug 2023 – Oct 2023, 3 months

- Crawled and preprocessed text data from diverse internet sources.
- Performed text mining using TF-IDF, Latent Dirichlet Allocation, and network analysis.

Daejeon Sejong Research Institute, External Technical Consultant — Daejeon, South Korea

Technical consulting for social media data collection and text mining. — July 2022 – Sept 2022, 3 months

- Advised on Python-based web crawling and preprocessing of crawled text data.
- Consulted on Latent Dirichlet Allocation and network analysis.

Korea Environment Institute, Part-time Modeler — Sejong, South Korea

Developed a deep learning-based spatial interpolation model for satellite-based CO_2 observations. — Aug 2021 – Oct 2021, 3 months

- Combined GEOS-Chem CO_2 modeling outputs and OCO-2 satellite data.
- Achieved stronger results than traditional interpolation methods.

Korea Research Institute for Human Settlements, Assistant Research Fellow — Sejong, South Korea

Spatial analysis and indicator development for national territorial monitoring. — Sept 2018 – Sept 2019, 1 year 1 month

- Built grid-based accessibility indicators to daily infrastructure such as schools and parks in South Korea.
- Created old-building proportion data using national building trajectory data.

Kyungpook National University, Student Research Assistant — Daegu, South Korea

Supported research on greenhouse gas satellite data and forest carbon offset assessment. — Nov 2015 – June 2016, 8 months

- Explored urban carbon distribution in Seoul using OCO-2 satellite data.
- Worked on Python-based satellite data preprocessing and geostatistical analysis.

Publications

Development of Grid-Based Population Projection Method: A Modified Cohort Component Approach Applied to South Korea — Jan 2025

Bokyeong Lee, *Byeonghwa Jeong*

10.1002/psp.70161 (Population, Space and Place)

A Large Language Model-based Chatbot System Framework for Urban Planners — Jan 2025

Xiaoxin Zhou, *Byeonghwa Jeong*, Karen Chapple

10.1016/j.eswa.2025.130307 (Expert Systems With Applications)

Popnet: computer vision based bespoken deep learning model for forecasting gridded population — Jan 2025

Byeonghwa Jeong, Bokyeong Lee

10.1080/13658816.2025.2514792 (International Journal of Geographical Information Science)

A first open retail boundary dataset for South Korea using open data and computer vision technique — Jan 2025

Byeonghwa Jeong, Juhwan Song, Younghoon Kim

10.1038/s41597-025-04958-1 (Scientific Data)

Does Class Matter? Understanding Differential Pandemic Recovery via a Building Typology Jan 2025
Byeonghwa Jeong, Karen Chapple
10.1177/23998083251326128 (Environment and Planning B: Urban Analytics and City Science)

Assessing Downtown Recovery Rates and Determinants in North American Cities After the COVID-19 Pandemic Jan 2024
Amir Forouhar, Karen Chapple, Jeff Allen, *Byeonghwa Jeong*, Julia Greenberg
10.1177/00420980241270987 (Urban Studies)

A new commercial boundary dataset for metropolitan areas in the USA and Canada, built from open data Jan 2024
Byeonghwa Jeong, Jeff Allen, Karen Chapple
10.1038/s41597-024-03275-3 (Scientific Data)

Topic Modeling of the Transition of the Sense of Place in Urban Regeneration Area: A Case Study of Daegu Bangcheon Market Jan 2020
Byeonghwa Jeong, Junwoo Kim
(Journal of Daegu Gyeongbuk Studies)

Application and Evaluation of Self-Organising Map-based Spatial Clustering for Regional Policy Recommendations Jan 2019
Byeonghwa Jeong, Junwoo Kim
(Journal of the Korean Geographical Society)

Comparative evaluation of South versus North Goseong County in terms of forest carbon offset potential Jan 2016
Byeonghwa Jeong, Jung-Sup Um
(Spatial Information Research)

Reports

- PopGNN: Graph Neural Network-Based Flexible Future Population Forecasting Model (2026). Byeonghwa Jeong, Viswadeep Lebakula, and Marie Urban. Oak Ridge National Laboratory.
- Satellite Embedding-Based Population Imputation for Areas with Missing Building Footprint Data (2026). Byeonghwa Jeong, Daniel Adams, Andrew Zimmer, and Marie Urban. Oak Ridge National Laboratory.
- Assessing the Doom Loop (2025). Karen Chapple, Amir Forouhar, Byeonghwa Jeong, and Stephanie Ortynsky. Lincoln Institute of Land Policy.
- Big Data-Driven Strategies for Enhancing the Science City Daejeon Brand (2025). Hyejin Joo, Sangheon Han, Eunjeong Jeong, Byeonghwa Jeong, Juhwan Song, and Minyoung Noh. Daejeon Sejong Research Institute.
- Basic Research for Citizen Participatory Urban Branding (2024). Hyejin Joo, Minyoung Noh, and Byeonghwa Jeong. Daejeon Sejong Research Institute.
- Discover Regional Things using Big Data Analysis and Civic Collaboration (2023). Hyejin Joo, Sangheon Han, Seolmin Yoon, Jaeyeon Moon, and Byeonghwa Jeong. Daejeon Sejong Research Institute.
- Development and Utilization of Multiple Deprivation Index for Place-based Living Facilities Policy (2021). Korea Research Institute for Human Settlements.
- National Territorial Monitoring Report in 2018 (2019). National Geographic Information Institute.

Work in Preparation

- Dual-Structure Spatio-Temporal Graph Learning for Hurricane-Induced Power Outage Forecasting. Byeonghwa Jeong, Sangkeun Lee, and Aaron Myers. In preparation for submission to AAAI AI for Social Impact.
- PopGNN: Graph Neural Network-Based Flexible Future Population Forecasting Model. Byeonghwa Jeong, Viswadeep Lebakula, and Marie Urban. In preparation for submission to AAAI AI for Social Impact.
- Building-Level Wealth Index Prediction Using DHS Survey Data and Machine Learning. Byeonghwa Jeong, Joe Tuccillo, and James Gaboardi. In preparation for submission to Computers, Environment and Urban Systems.
- Satellite Embedding-Based Population Imputation for Areas with Missing Building Footprint Data. Byeonghwa Jeong, Andrew Zimmer, and Daniel Adams. In preparation for submission to ACM Knowledge Discovery and Data Mining.
- Global Level Livability Mask for Gridded Population Datasets. Byeonghwa Jeong, Viswadeep Lebakula, and Brandyn Reynolds. In preparation for submission to Scientific Data.
- Framework for Rapid Updates to LandScan High-Resolution Gridded Population Datasets. Andrew Zimmer, Daniel Adams, Byeonghwa Jeong, and Marie Urban. In preparation for submission to Scientific Reports.

Conference Presentations

- Building-Level Wealth Index Prediction Using DHS Data and Machine Learning (2026). Byeonghwa Jeong, Joe Tuccillo, and James Gaboardi. AAG Annual Meeting.
- Delineating commercial boundaries using an AI-based computer vision technique with open data in South Korea (2024). Young-Hoon Kim and Byeonghwa Jeong. AsiaCarto 2024.
- Typology of Downtown Recovery at the Granular Scale (2024). Byeonghwa Jeong, Karen Chapple, and Jeff Allen. Geoinformatics 2024.

Grants

- A Study on the Evaluation and Characteristics of Coastal Area by Using Image Segmentation Model. Korea Maritime Institute. Award amount: USD 2,885. Personal contribution: idea development, core data provision, writing, and review.
- Assessing the Doom Loop: Scenarios in Four North American Downtowns. Lincoln Institute of Land Policy. Award amount: USD 52,931. Personal contribution: data curation, data collection, preprocessing, explainable AI, and proposal writing.
- Marshalling a Robust Downtown Recovery for Ottawa. Canadian Urban Institute. Award amount: USD 109,514. Personal contribution: data curation, data collection, preprocessing, methodology selection, analysis, visualization, and proposal writing.

Teaching and Supervision

- Graduate student supervision (2024). Xiaoxin Zhou, M.S. in Informatics, University of Toronto; Gaweon Shin, Ph.D. in Public Administration, Yonsei University; Changwha Oh, Ph.D. in Geography, University of Tennessee Knoxville.
- Undergraduate student supervision (2024). Christoper Kim, B.A., UC Berkeley; Brandon Choi, B.A., UC Berkeley.
- Guest lecture (2025). The application of computer vision for delineating regional boundary, Korea Research Institute for Human Settlements.
- Guest lecture (2024). The application of AI for social phenomenon prediction, Korea National University of Education.
- Guest lecture (2023). Social phenomenon and spatial AI, Kyungpook National University.

Service

- Conference and event management. Member of the Organising Committee, International Conference on Geospatial Information Science, Spatially Enabled Society with AI and Digital Twin, 2019.
- Journal reviewer. International Journal of Geographical Information Science, Taylor & Francis, 2025.
- Journal reviewer. Computers and Electrical Engineering, Elsevier, 2025.
- Journal reviewer. Papers in Regional Science, Elsevier, 2026.
- Journal reviewer. Spatial Information Research, Springer, 2026.
- Journal reviewer. Transactions in GIS, Wiley, 2026.

Honours and Prizes

- Best Paper Award (2020). Awarded by Daegu Gyeongbuk Development Institute for research on sense of place in an urban regeneration area.
- Outstanding Paper Award (2015). Awarded by the Korea Spatial Information Society.
- June-Myeong Scholarship (2013). Awarded by GEO C&I Corp. for skilled ArcGIS use.

Technical Skills

GIS analysis: ArcGIS, QGIS

Programming: Python

Data query and management: Pandas, GeoPandas, NumPy, PySpark

Machine learning and deep learning: PyTorch, Torchvision, scikit-learn, Gensim

Statistics: Regression, multivariate multiple regression, likelihood ratio tests, correlation tests, feature engineering

Web crawling and text mining: Selenium, BeautifulSoup, Requests, TF-IDF, Word2Vec, Latent Dirichlet Allocation

Large language models: Prompt engineering, fine-tuning

Network analysis: NetworkX, UCINET

Remote sensing: Google Earth Engine, Python-based raster analysis

Cloud and databases: Google Firebase, Google Colab, MariaDB

Languages

Korean: Native speaker

English: Fluent